MATH 3312, SPRING 2026: HOMEWORK 2

(1) Determine if the following statements are true or false. Give a line or two in justification.
   (a) Every finite integral domain is a field.
   (b) There exists a finite field of order 24.
   (c) There exists a finite field of order 8.
   (d) There are no non-trivial ring homomorphisms
$$\mathbb{Z}[x]/(x^2+1) \to \mathbb{F}_7.$$
   (e) There exists a finite field k with $k^\times \simeq \mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/2\mathbb{Z}$.

(2) Give examples of the following:
   (a) A finite field of order 49.
   (b) A generator for $\mathbb{F}_{11}^\times$.
   (c) A finite field with no elements of order 3.

(3) Let k be a finite field of characteristic p (so that it contains $\mathbb{F}_p$).
   (a) Show that, for all elements $\alpha, \beta \in k$, we have
$$(\alpha+\beta)^p = \alpha^p + \beta^p.$$
   (b) Show that $1 \in k$ is the only zero for $x^p - 1 \in k[x]$.
   (c) Show that the function $\mathrm{Fr} : \alpha \mapsto \alpha^p$ is a ring *automorphism* from k to itself, and that $\mathrm{Fr}(a) = a$ for all $a \in \mathbb{F}_p$.

(4) Show that the following are equivalent for a field k of order p^d (where p is prime):
   (a) $k^\times$ contains an element of order n.
   (b) $p^d \equiv 1 \pmod{n}$.

(5) Let k be a field of order p^d. Show that every element $a \in k$ is a zero for the polynomial $x^{p^d} - x \in \mathbb{F}_p[x]$.

(6) Show that the polynomial x^2+1 is irreducible in $\mathbb{F}_p[x]$ if and only if $p \equiv 3 \pmod{4}$.

(7) Show that the polynomial x^2+x+1 is irreducible in $\mathbb{F}_p[x]$ if and only if $p \equiv 2 \pmod{3}$. Can you come up with a similar criterion for $x^2 - x + 1$?